



香港科技大學
THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY

ELEC 2100 Signals and Systems

Wenchao Miao



Outline Chapter 4 Frequency domain analysis of LTIS



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AND TECHNOLOGY

1 Overview

2 Frequency Response of System

3 Amplitude Modulation

4 Sampling Theorem and Aliasing

5 Fourier Transform of a Sequence (DTFT)

6 Discrete Time Fourier Series

7 DTFT and DFT

8 Fast Fourier Transform (Reference)

1. Overview of Frequency Domain Analysis OF LTIS

This chapter preliminarily studies several aspects of **applying Fourier transform method to communication systems**

Fourier Transform Property	Application
Time-domain Convolution Theorem	Filtering , System Frequency Response Characteristics
Frequency Shift Property	Modulation , Frequency Division Multiplexing
Frequency-Domain Convolution Theorem	Sampling , Time Division Multiplexing

1. Overview of DTFT

From **Continuous-Time** Fourier Transform (CTFT) to **Discrete-Time** Fourier Transform (DTFT)

- Discrete Time Fourier Series (DTFS)
- Properties of Discrete Time Fourier Series
- Discrete Time Fourier Transform (DTFT)
- Properties of Discrete Time Fourier Transform
- Fast Fourier Transformer (FFT)

2. Frequency Response of System

Main Content

Fourier Transform Analysis Method:

- System Function in the Form of Fourier Transform
- Ideal Low Pass Filter

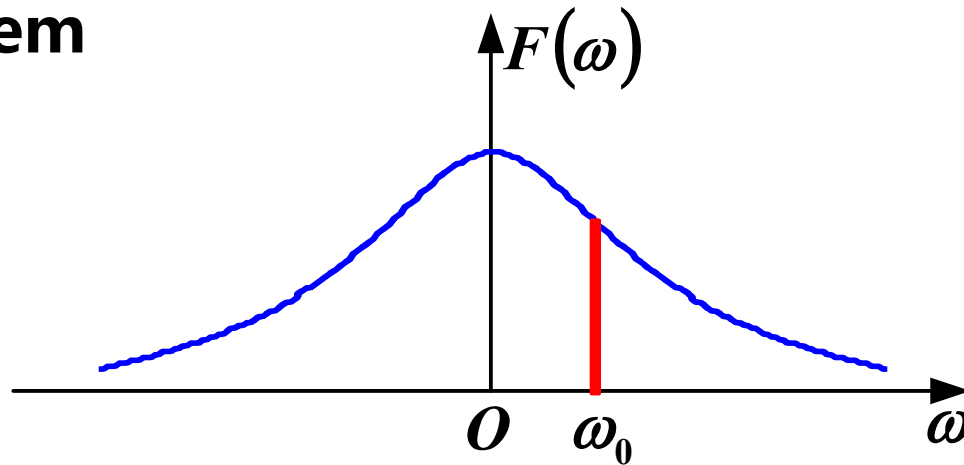
2. Frequency Response of System

1. System Function in the Form of Fourier Transform

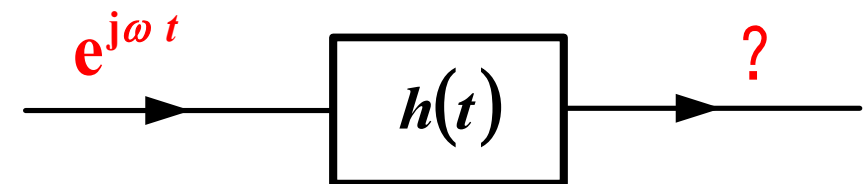
Decomposition: A signal can be decomposed into fundamental signals $e^{j\omega t}$

Response Calculation: Determine the response of the fundamental signal $e^{j\omega t}$ passing through the system

Superposition: The response of the signal passing through the system

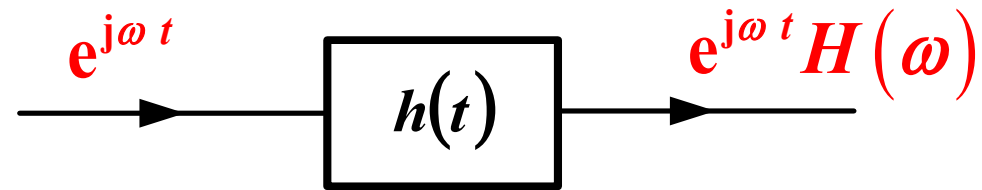


$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega$$



2. Frequency Response of System

(1) Response of the fundamental signal $e^{j\omega t}$ passing through the system



System Response

$$\begin{aligned} r_1(t) &= e^{j\omega t} * h(t) = \int_{-\infty}^{\infty} h(\tau) e^{j\omega(t-\tau)} d\tau \\ &= e^{j\omega t} \int_{-\infty}^{\infty} h(\tau) e^{-j\omega\tau} d\tau \\ &= e^{j\omega t} H(\omega) \end{aligned}$$

Where

$$H(\omega) = \mathcal{F}[h(t)] = \int_{-\infty}^{\infty} h(\tau) e^{-j\omega\tau} d\tau$$

2. Frequency Response of System

(2) Signals can be expressed as a sum of fundamental signals

Periodic Signals

$$e(t) = \sum_{k=-\infty}^{\infty} E_k e^{jk\omega_1 t}$$

Fundamental Signal $e^{jk\omega_1 t}$

Aperiodic Signals

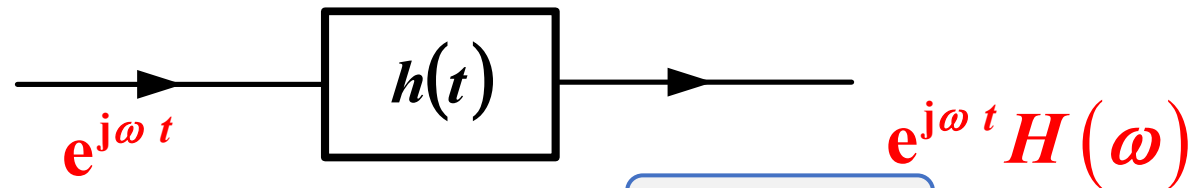
$$e(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} E(\omega) e^{j\omega t} d\omega$$

Fundamental Signal $e^{j\omega t}$

2. Frequency Response of System

(3) Response of an arbitrary signal through a system

Method: Decompose the signal, determine individual responses, then superimpose



Component Response:

$$\frac{1}{2\pi} E(\omega) d\omega e^{j\omega t}$$

$$\frac{1}{2\pi} E(\omega) d\omega e^{j\omega t} H(\omega)$$

Homogeneity

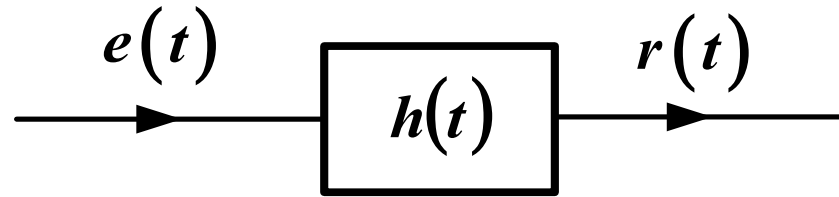
$$e(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} E(\omega) e^{j\omega t} d\omega \quad r(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} E(\omega) e^{j\omega t} H(\omega) d\omega$$

Additivity

$$R(\omega) = E(\omega) \cdot H(\omega)$$

2. Frequency Response of System

(3) Response of an arbitrary signal through a system



System Response

$$r(t) = e(t) * h(t)$$

Let

$$\mathcal{F}[e(t)] = E(\omega), \mathcal{F}[h(t)] = H(\omega), \mathcal{F}[r(t)] = R(\omega)$$

Using the time-domain convolution theorem, we obtain

$$R(\omega) = E(\omega) \cdot H(\omega)$$

2. Frequency Response of System

(4) System function in the Fourier transform domain

$$H(\omega) = \frac{R(\omega)}{E(\omega)} \quad \text{or } H(j\omega)$$

$H(\omega)$ is referred to as the **system function** in the Fourier transform domain

$H(\omega) \sim \omega$ is also termed the **frequency response characteristic** of the system

$$H(\omega) = |H(\omega)| e^{j\varphi(\omega)}$$

$|H(\omega)| \sim \omega$ **Magnitude-Frequency Characteristic**

$\varphi_h(\omega) \sim \omega$ **Phase-Frequency Characteristic**

2. Frequency Response of System

(5) Physical interpretations of the system function

	$R(\omega) = E(\omega) \cdot H(\omega)$	
Excitation	$E(\omega) = E(\omega) \cdot e^{j\varphi_e(\omega)}$	frequency-dependent weighting on the signal's spectral components
System Function	$H(\omega) = H(\omega) \cdot e^{j\varphi_h(\omega)}$	
Response	$R(\omega) = R(\omega) \cdot e^{j\varphi_r(\omega)}$	

Through complex function operations, we obtain

$$|R(\omega)| = |E(\omega)| \cdot |H(\omega)|$$

The amplitude $E(\omega)$
is scaled by $|H(\omega)|$

$$\varphi_r(\omega) = \varphi_e(\omega) + \varphi_h(\omega)$$

The phase of $E(\omega)$
is shifted by $\varphi_h(\omega)$

2. Frequency Response of System

(6) Methods for determining the system function

- **From Differential Equations**
- **From Circuit Analysis**
- **From Impulse Response**

2. Frequency Response of System

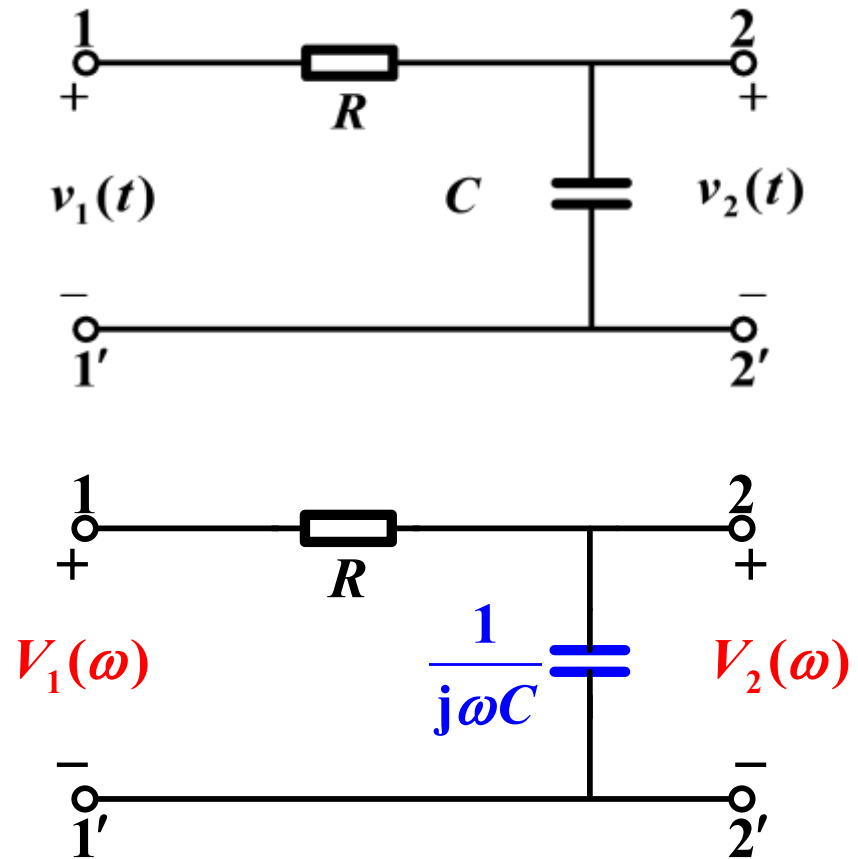
Example 1

Determine the system function $H(\omega)$ for the given RC circuit

Represent the circuit in the frequency domain and derive the system function

$$H(\omega) = \frac{V_2(\omega)}{V_1(\omega)} = \frac{\frac{1}{j\omega C}}{R + \frac{1}{j\omega C}}$$

Let $\alpha = \frac{1}{RC}$, $H(\omega) = \frac{\alpha}{\alpha + j\omega}$



2. Frequency Response of System

Example 1

Frequency Response Characteristics of the System

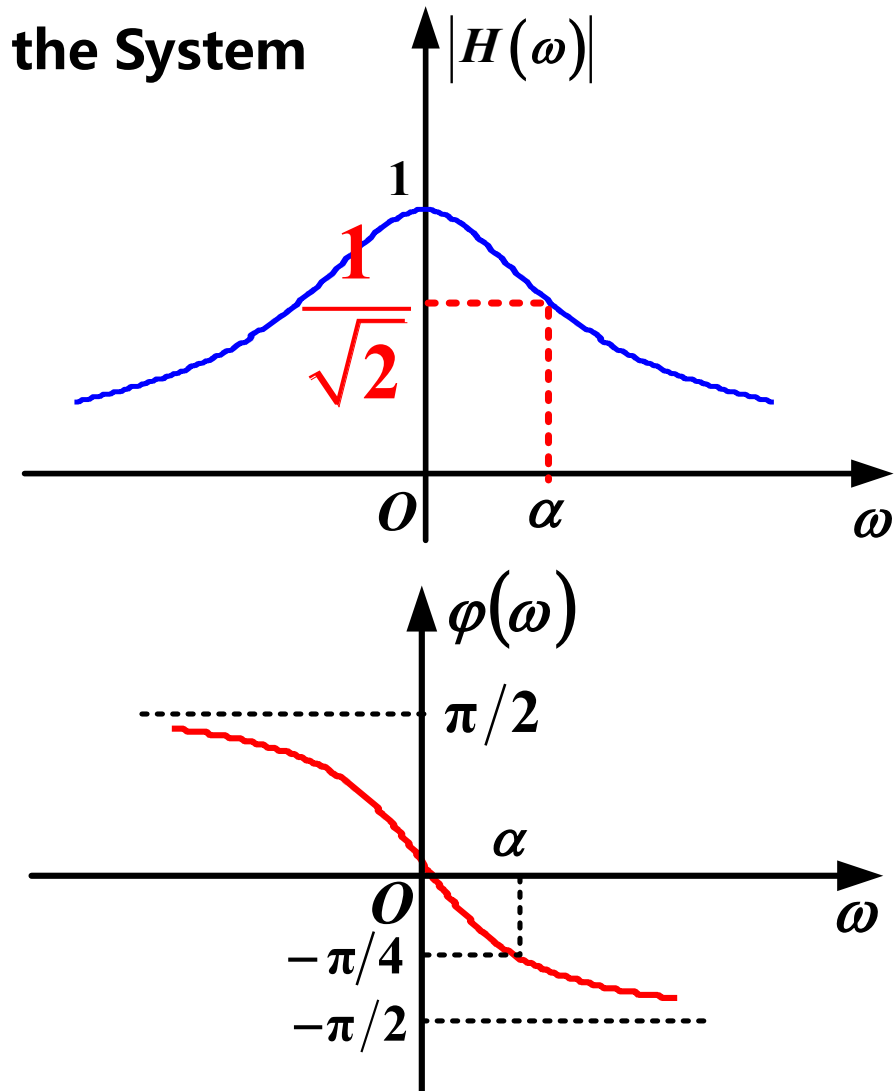
$$H(\omega) = \frac{\alpha}{\alpha + j\omega} \quad \left(\alpha = \frac{1}{RC} \right)$$

$$|H(\omega)| = \frac{\alpha}{\sqrt{\alpha^2 + \omega^2}}$$

$$\varphi(\omega) = -\arctan \frac{\omega}{\alpha}$$

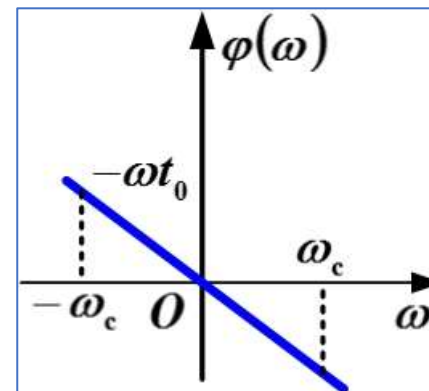
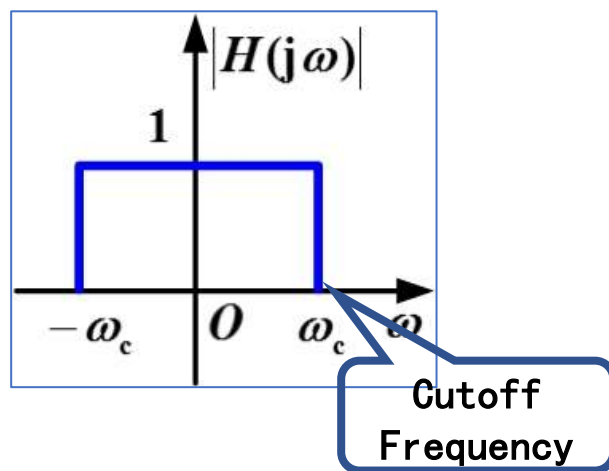
Low-Pass Filter

The **cutoff frequency** is determined by α
 α corresponds to the **3dB** bandwidth
(frequency at which power is halved)



2. Frequency Response of System

2. Ideal Low Pass Filter—Frequency Domain Characteristics



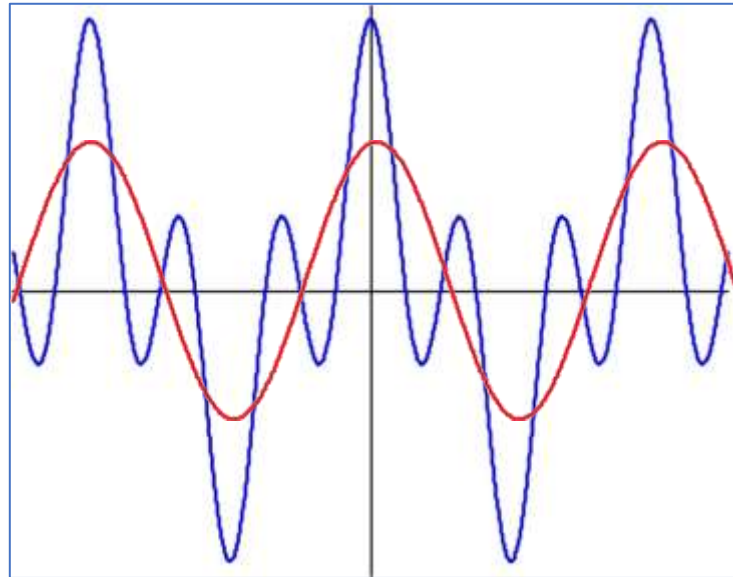
$$H(j\omega) = \begin{cases} 1 \cdot e^{-j\omega t_0} & |\omega| < \omega_c \\ 0 & |\omega| > \omega_c \end{cases} \Rightarrow \begin{cases} |H(j\omega)| = \begin{cases} 1 & |\omega| < \omega_c \\ 0 & |\omega| > \omega_c \end{cases} \\ \varphi(\omega) = -\omega t_0 \end{cases}$$

2. Frequency Response of System

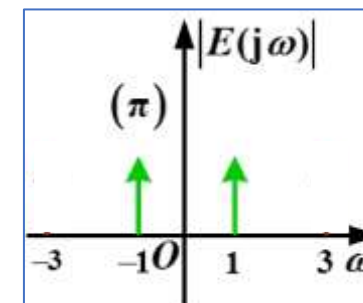
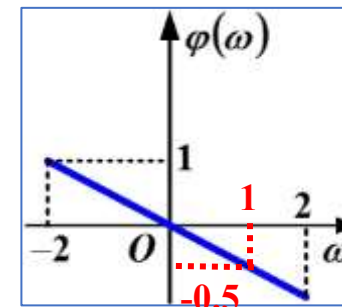
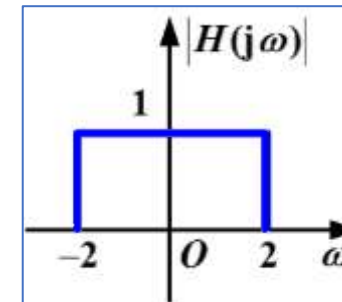
Example 2

An input signal $e(t) = \cos t + \cos 3t$ passes through an ideal low-pass filter $H(j\omega)$

Determine the output signal $r(t)$



$$r(t) = \cos\left(t - \frac{1}{2}\right)$$



2. Frequency Response of System

Summary

$$H(\omega) = \frac{R(\omega)}{E(\omega)}$$

- $H(\omega)$ **System Function in the Form of Fourier Transform**

$$|R(\omega)| = |E(\omega)| \cdot |H(\omega)|$$

The amplitude $E(\omega)$
is scaled by $|H(\omega)|$

$$\varphi_r(\omega) = \varphi_e(\omega) + \varphi_h(\omega)$$

The phase of $E(\omega)$
is shifted by $\varphi_h(\omega)$

- **Ideal Low Pass Filter**

Main Content

- **Modulation and Demodulation**
- **Introduction to Modulation Methods**
- **Amplitude Modulation**
- **Demodulation**
- **Frequency Division Multiplexing**
- **Time Division Multiplexing**

1. Modulation and Demodulation

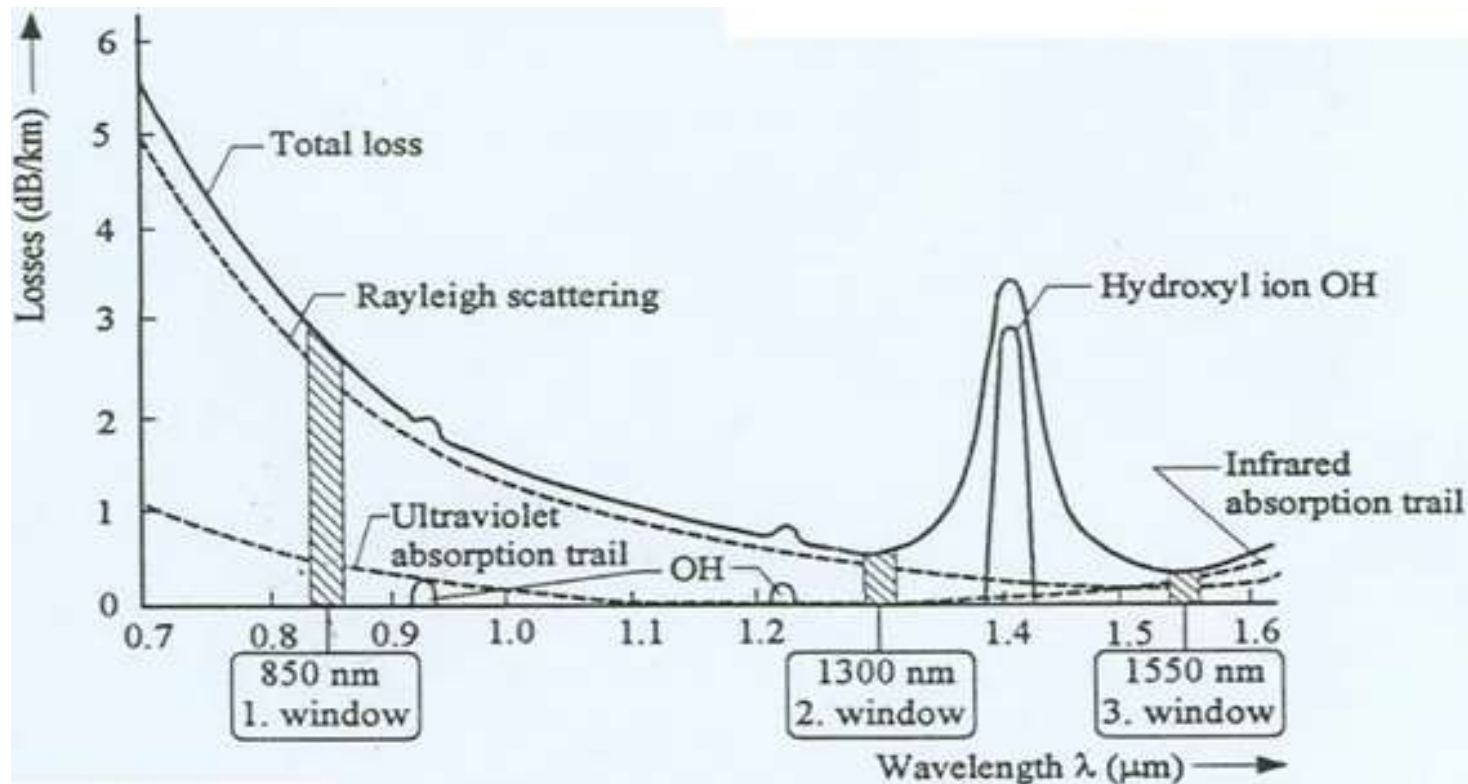
In a communication system, signals are transmitted from the transmitter to the receiver. To achieve signal transmission, **modulation and demodulation** are often required

- **Modulation:** The process of embedding an information-carrying signal into another signal
- **Demodulation:** The process of extracting the information-carrying signal from the modulated signal

3. Amplitude Modulation

(1) Why is Modulation Necessary?

To adapt the signal for transmission over a suitable channel



Fiber loss and working window

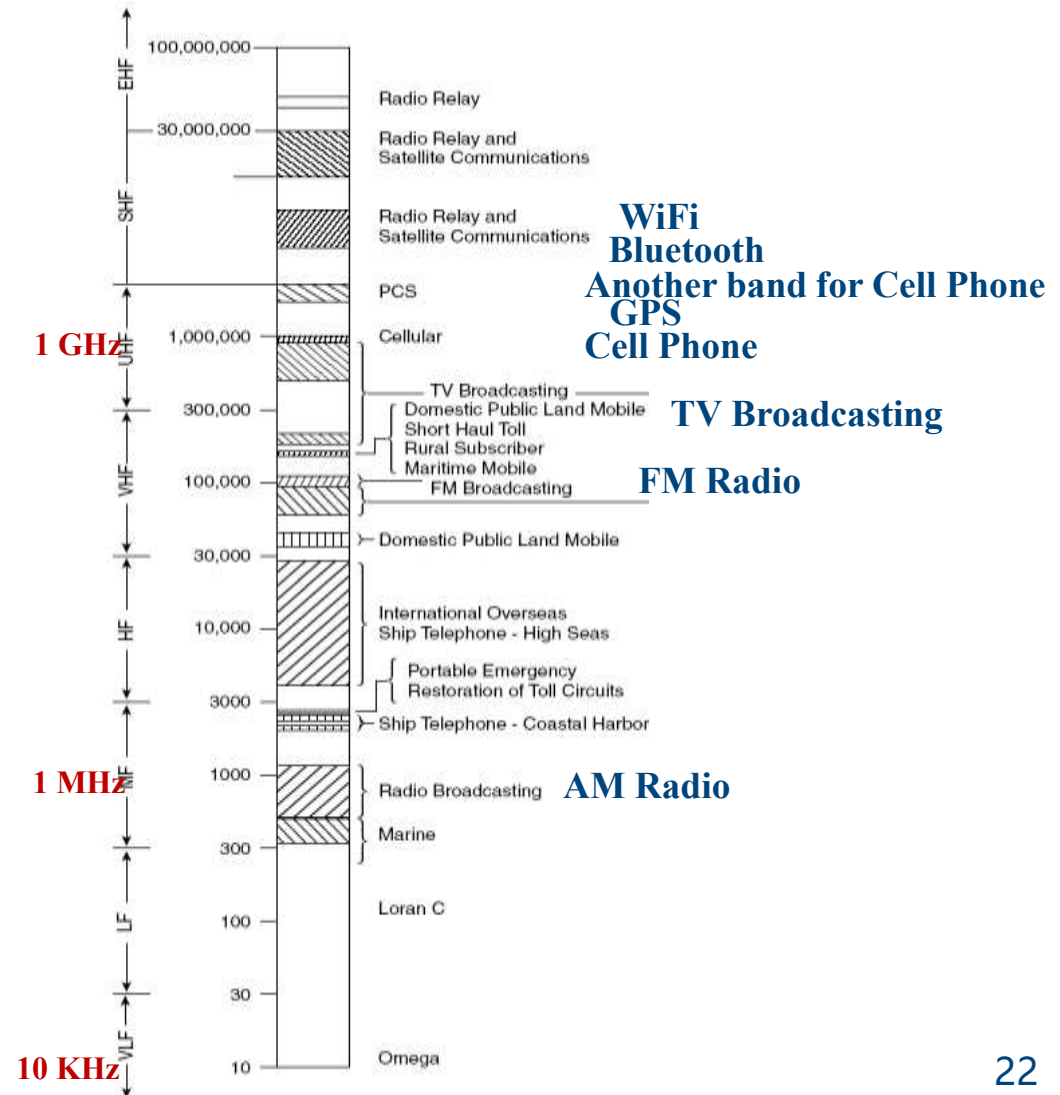
3. Amplitude Modulation

(1) Why is Modulation Necessary?

For about 120 years, human have used radio waves in the range of 10 KHz to 100 GHz for many communications applications other than AM and FM radio broadcasting: marine and aviation communications, terrestrial TV, Satellite TV, walkie talkie, cordless phone, car key, garage door opener, cell phone (2G, 3G, 4G, 5G), WiFi, Bluetooth, RFID, GPS..... etc.

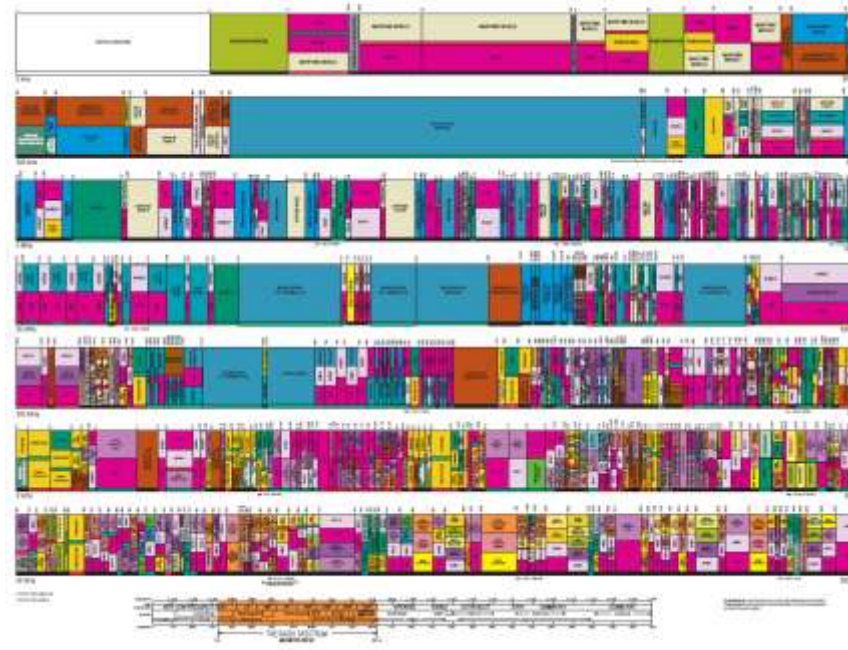
Radio spectrum is a valuable resource with usage governed by governments and international treaties. To the right is a high-level view of how the radio spectrum is allocated to different applications in the US.

Often one has to pay large sum of \$\$ for the use of specific spectrums – spectrum auction.



3. Amplitude Modulation

(1) Why is Modulation Necessary?

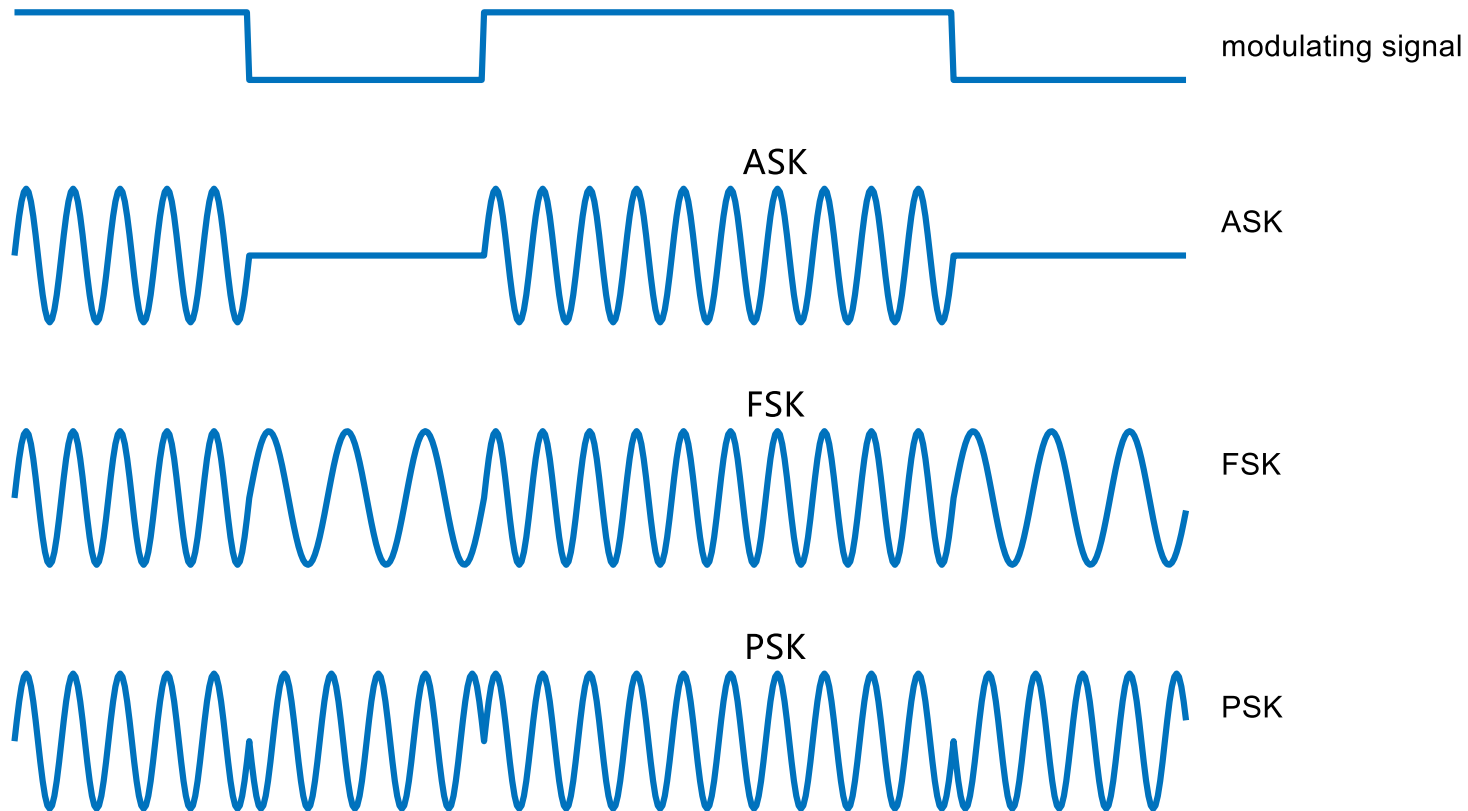


- The detailed frequency allocation table for different applications is extremely complex. It also varies from country to country
- Further, each application may have several to hundreds of channels (e.g., different TV/radio channels) sharing the allocated spectrum
- **Today, every bit of the usable radio spectrum is used!** Because spectrum is so valuable, different applications and different channels are often closely packed with little room between them. (*Hence we need good LPF or BPF to separate different channels.*)

3. Amplitude Modulation

2. Introduction to Modulation Methods

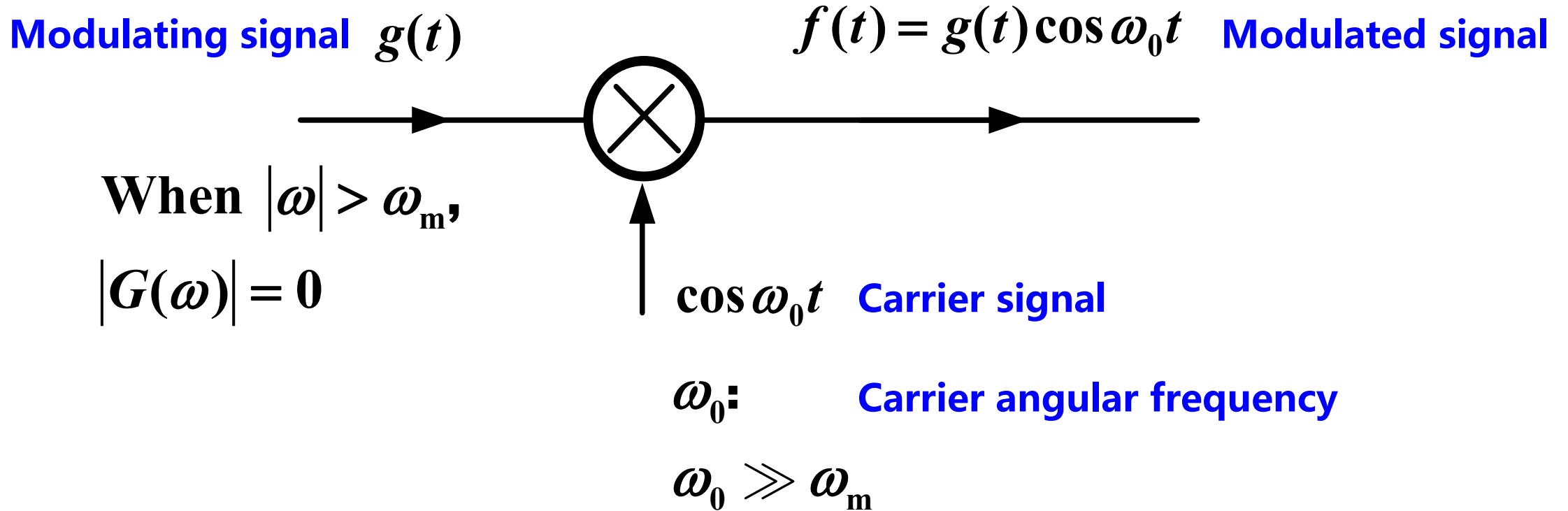
$$f(t) = A \sin(2\pi f t + \theta)$$



3. Amplitude Modulation

3. Amplitude Modulation (Suppressed-carrier amplitude modulation, AM-SC)

(1) System Block Diagram



(2) Spectral Structure

$$f(t) = g(t) \cdot \cos(\omega_0 t) \quad \underline{\text{Euler's formula}} \quad \frac{1}{2} g(t) \left[e^{j\omega_0 t} + e^{-j\omega_0 t} \right]$$

Modulation
theorem

$$F(\omega) = \frac{1}{2} \left[G(\omega - \omega_0) + G(\omega + \omega_0) \right]$$

$$f(t) = g(t) \cos(\omega_0 t)$$

$$\xleftarrow{\text{Convolution theorem}} \frac{1}{2\pi} G(\omega) * \left[\pi \delta(\omega - \omega_0) + \pi \delta(\omega + \omega_0) \right]$$

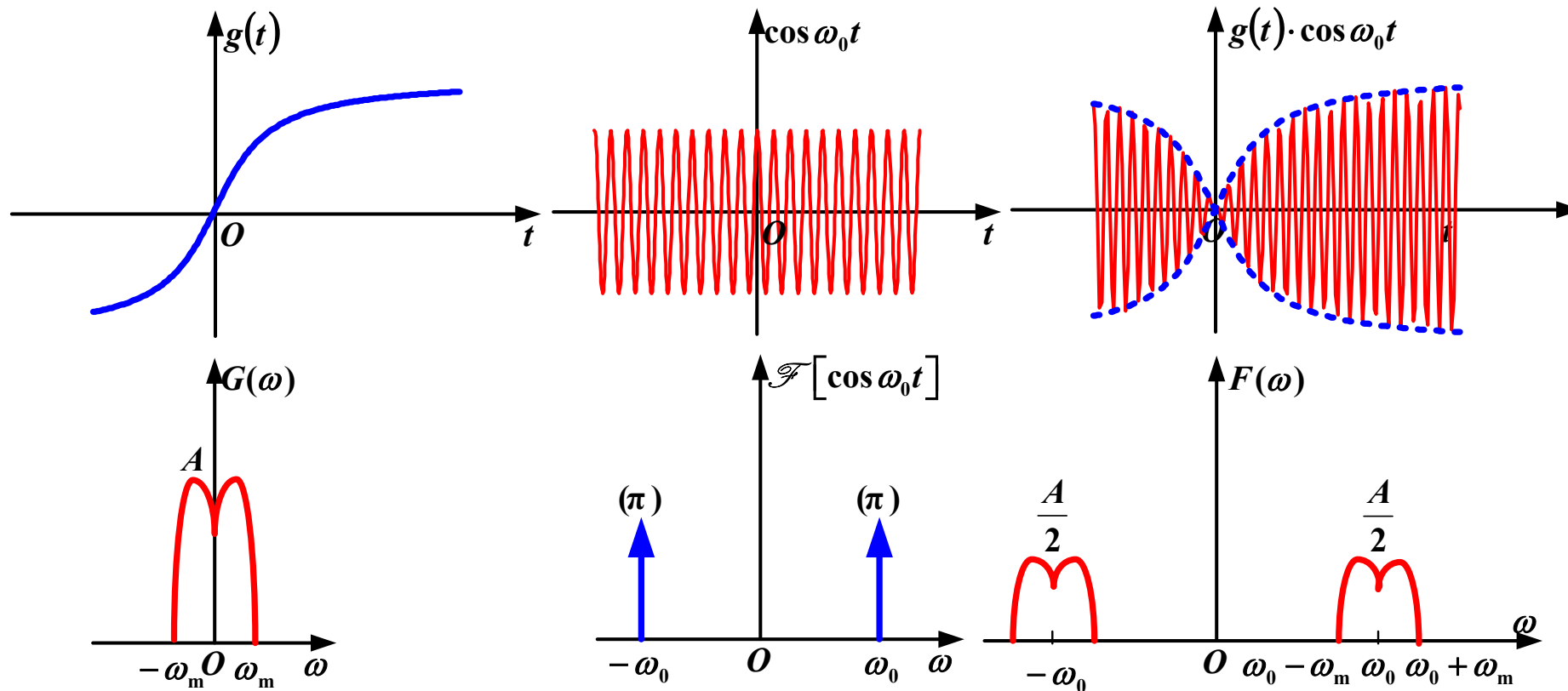
Frequency Shift

Convolution

3. Amplitude Modulation

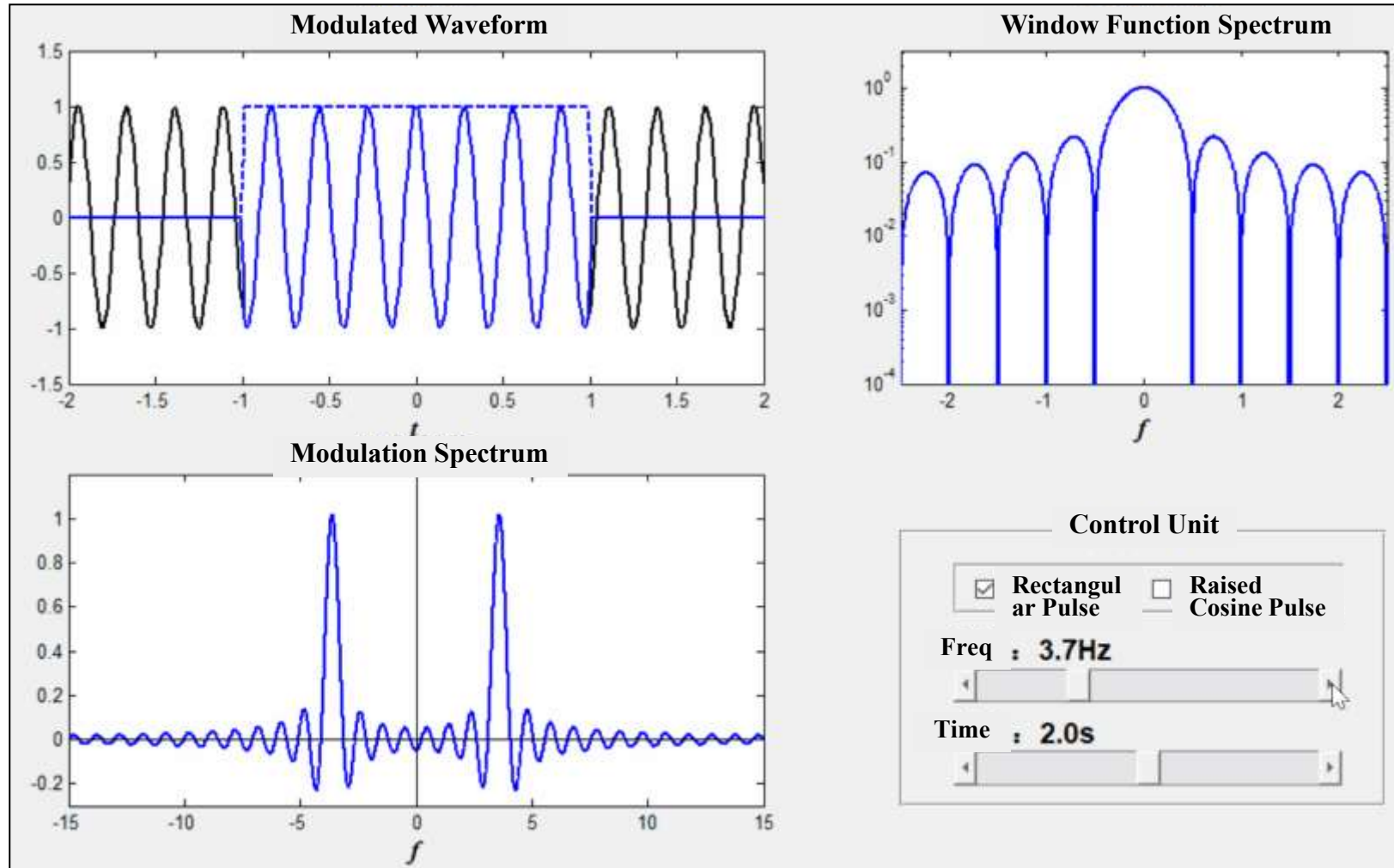
(2) Spectral Structure

$$f(t) = g(t) \cdot \cos(\omega_0 t) \quad F(\omega) = \frac{1}{2} [G(\omega - \omega_0) + G(\omega + \omega_0)]$$



3. Amplitude Modulation

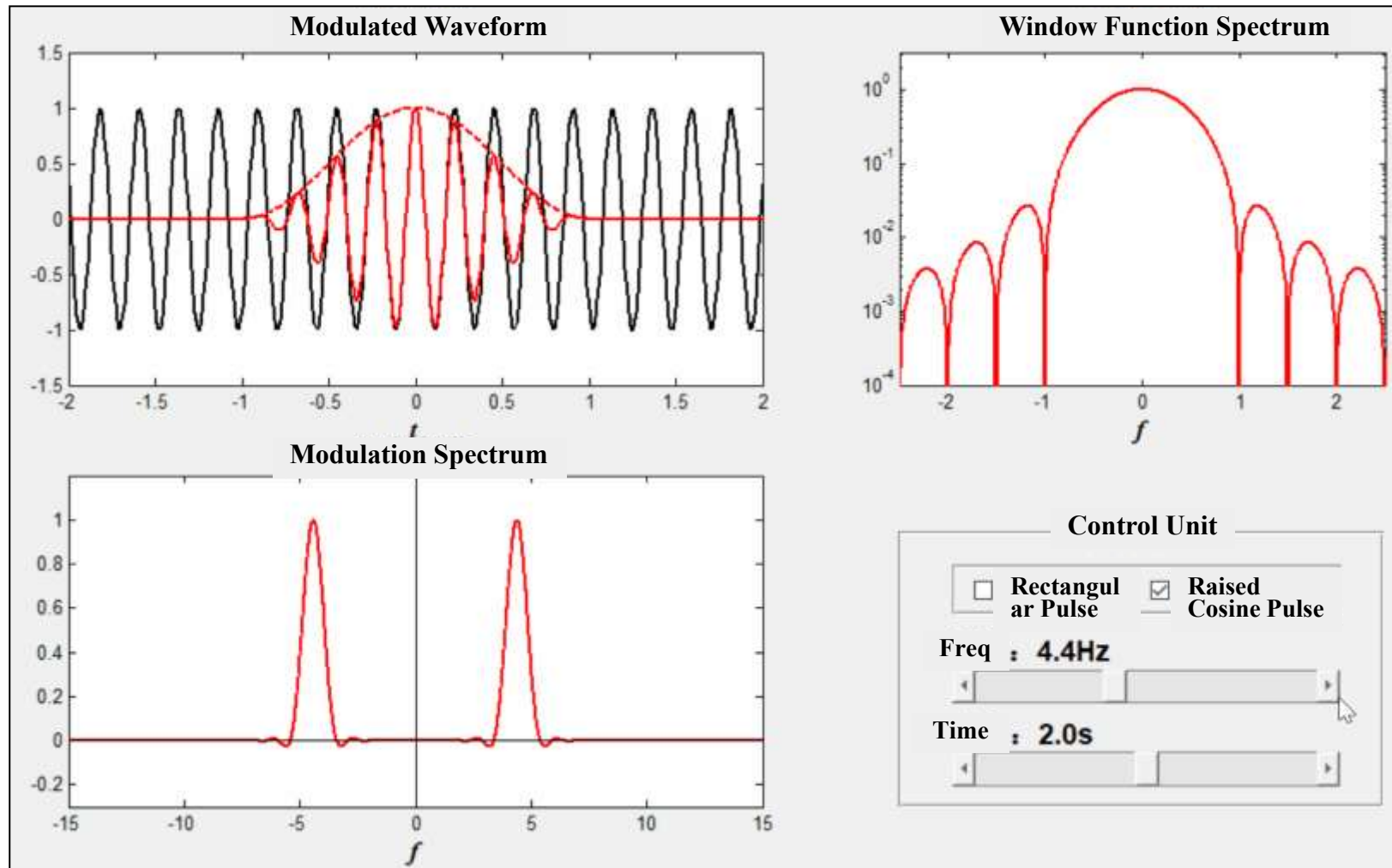
(2) Spectral Structure



3. Amplitude Modulation



(2) Spectral Structure

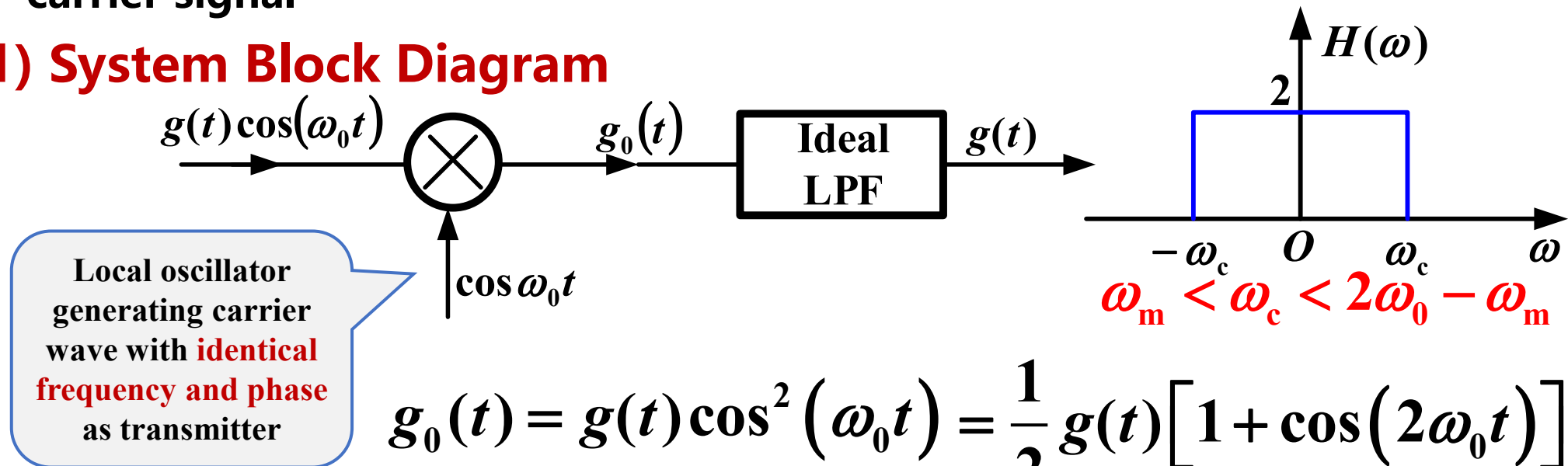


3. Amplitude Modulation

4. Demodulation (Coherent Detection/Synchronous Detection)

The process of recovering the original baseband signal from the modulated carrier signal

(1) System Block Diagram

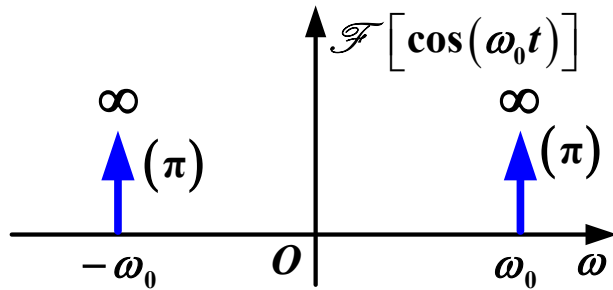
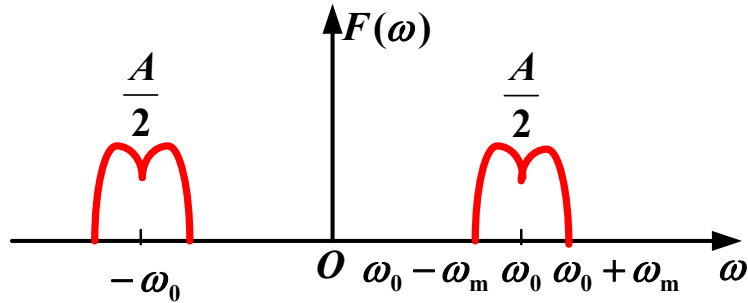


$$g_0(t) = g(t) \cos^2(\omega_0 t) = \frac{1}{2} g(t) [1 + \cos(2\omega_0 t)]$$

$$G_0(\omega) = \frac{1}{2} \mathbf{G(\omega)} + \frac{1}{4} G(\omega - 2\omega_0) + \frac{1}{4} G(\omega + 2\omega_0)$$

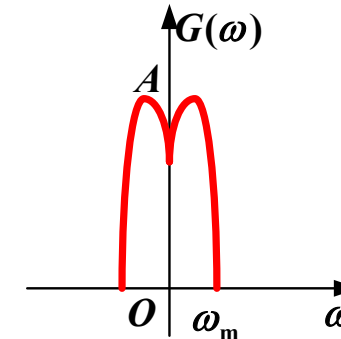
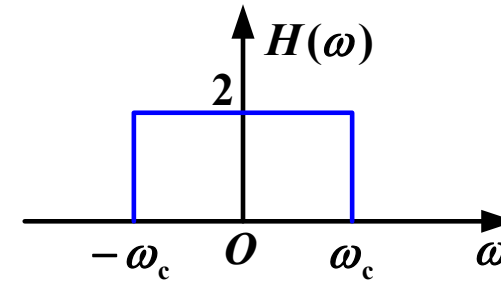
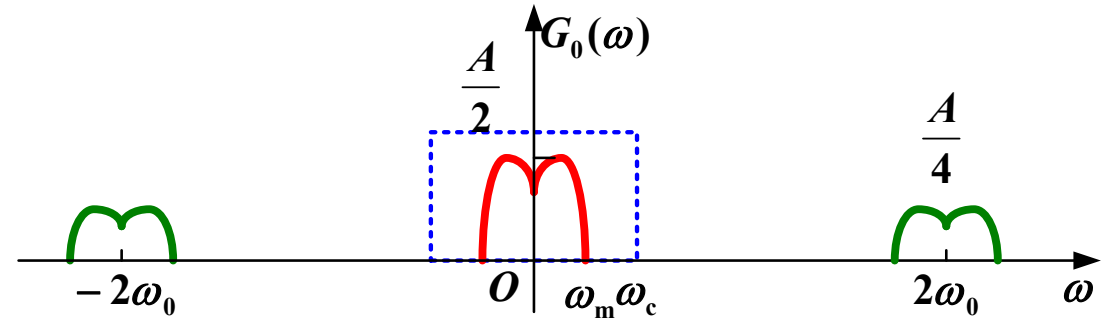
3. Amplitude Modulation

(2) Spectral Structure



$$G_0(\omega)H(\omega) = G(\omega)$$

$$\omega_m < \omega_c < 2\omega_0 - \omega_m$$



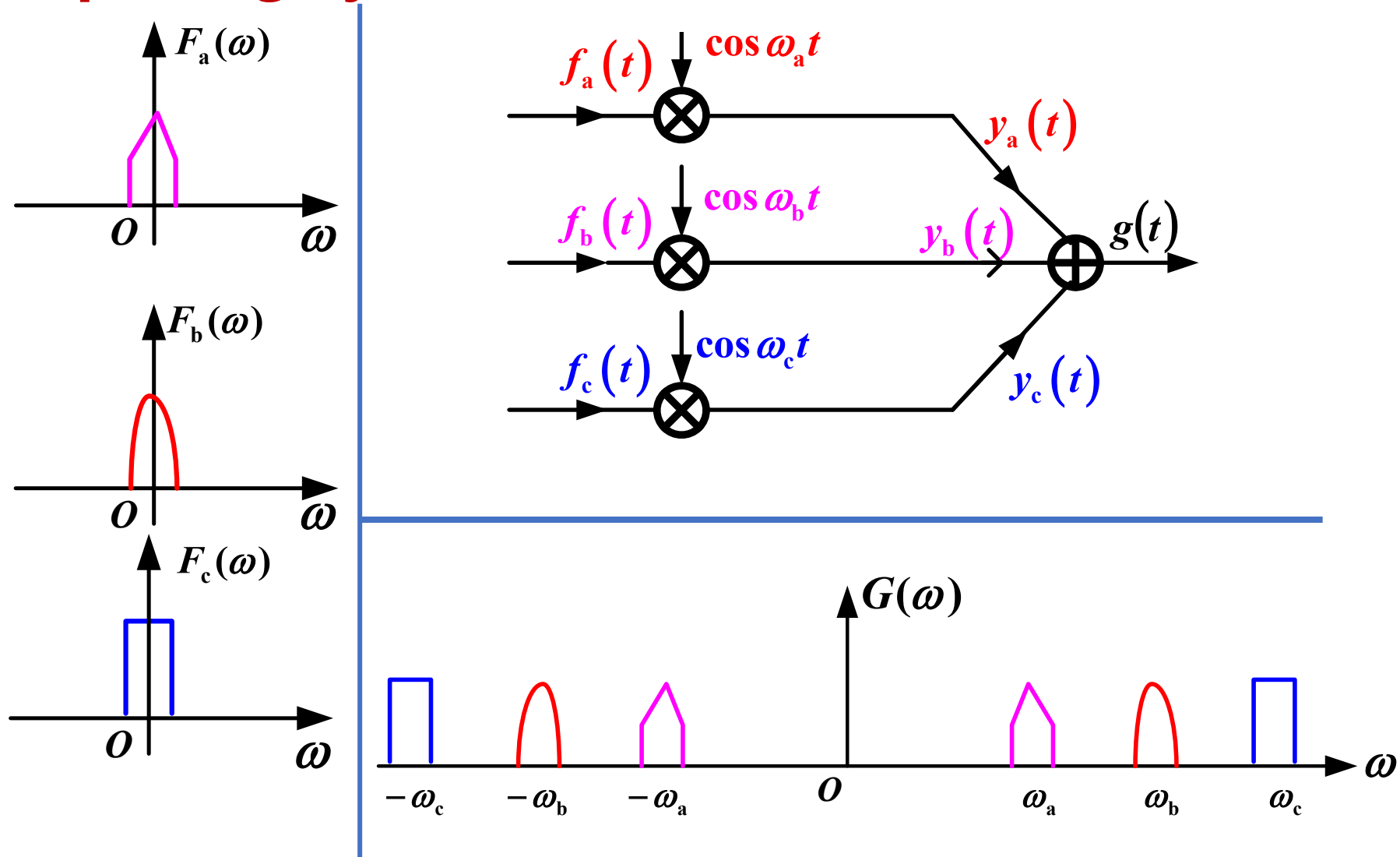
5. Frequency Division Multiplexing

(1) Definition

- Frequency Division Multiplexing (FDM)
- A transmission scheme that enables multiple communication channels within a single physical medium by dividing the available bandwidth into distinct frequency sub-bands.

3. Amplitude Modulation

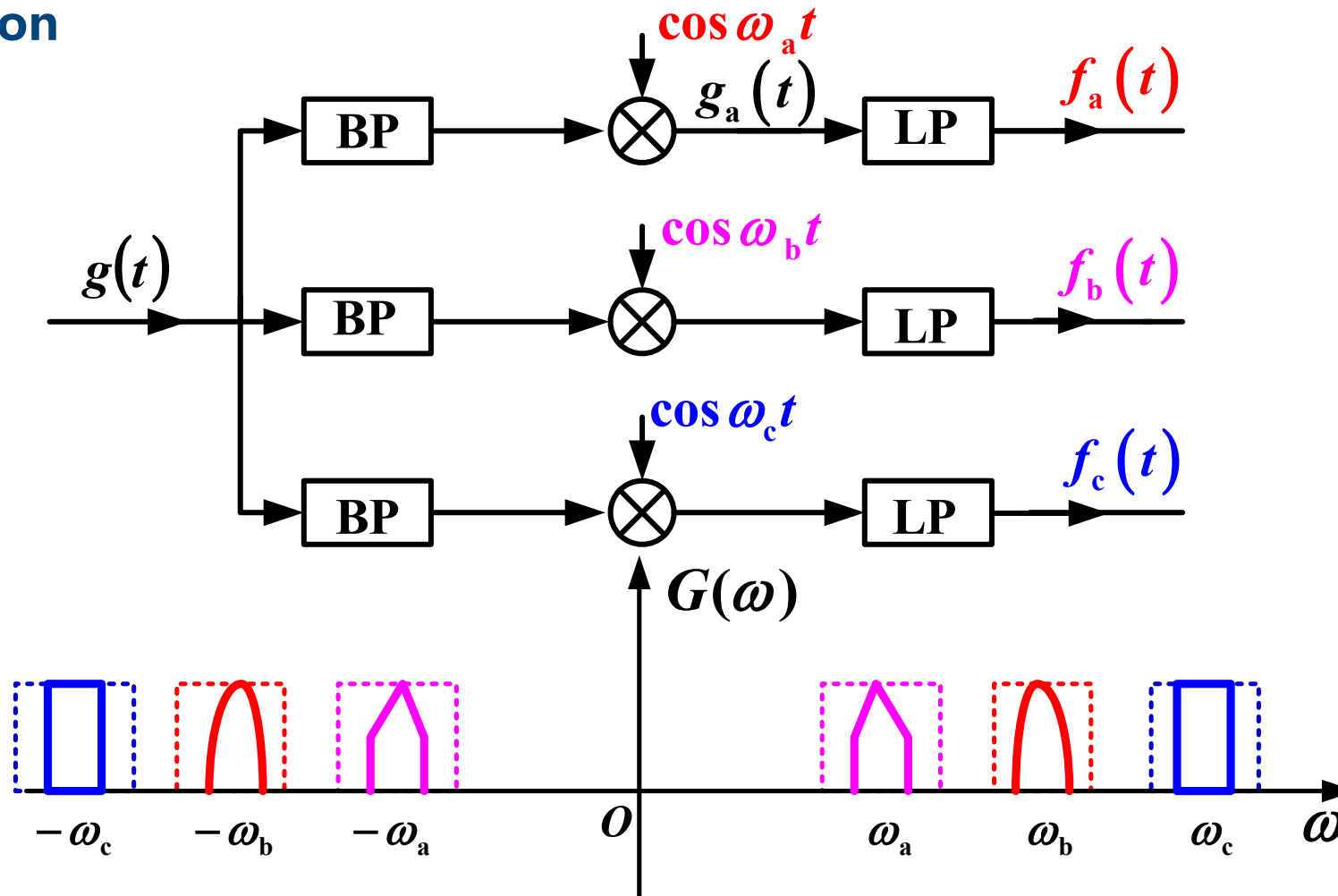
(2) Multiplexing System Structure



3. Amplitude Modulation

(3) Demultiplexing System Structure

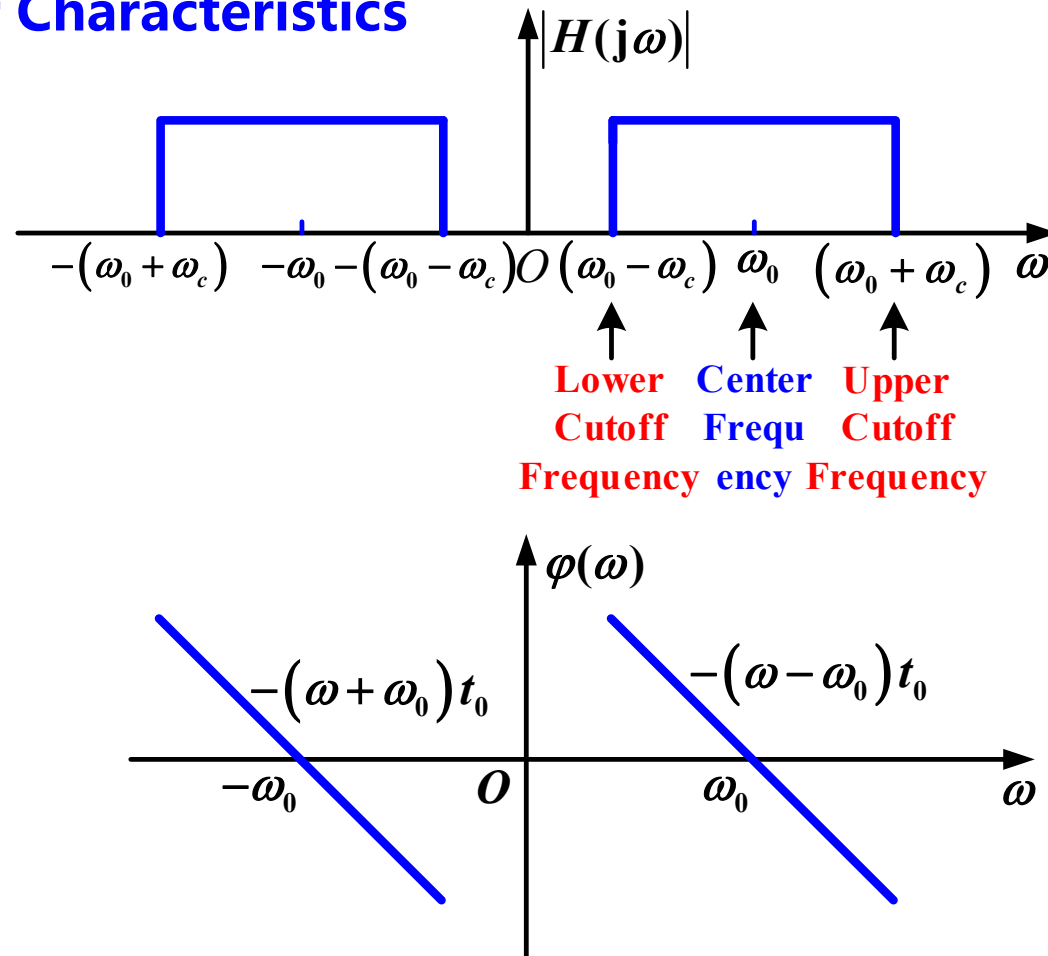
Uses **bandpass filters** to separate each channel signal, followed by individual demodulation



3. Amplitude Modulation

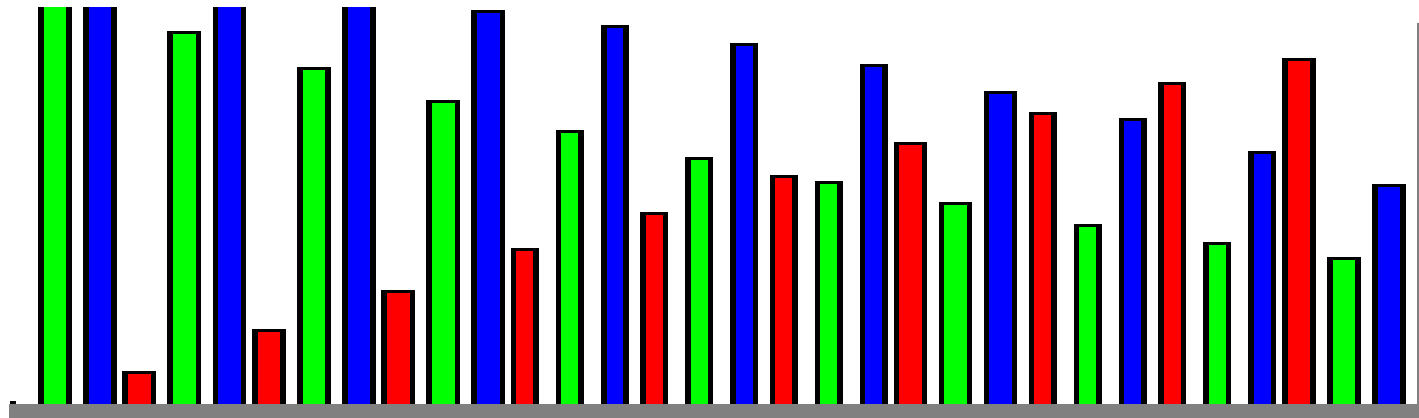
(3) Demultiplexing System Structure

Bandpass Filter Characteristics



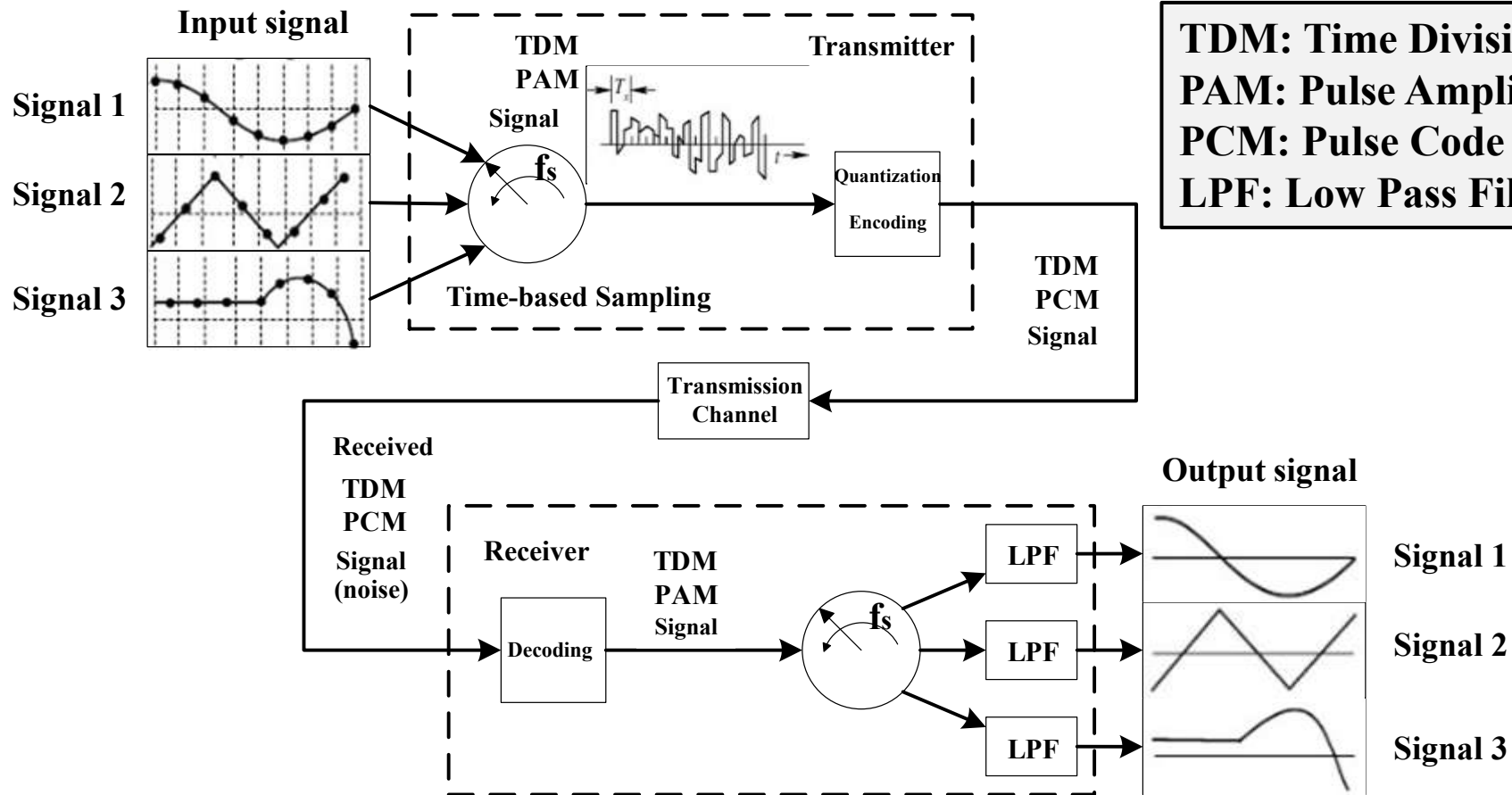
6. Time Division Multiplexing

- **Primary Application:** Digital signal transmission and access systems
- **Core Principle:** Divides the transmission channel into discrete time intervals
- **Key Terminology:** Time Slot, Each allocated segment of the transmission timeframe



3. Amplitude Modulation

6. Time Division Multiplexing

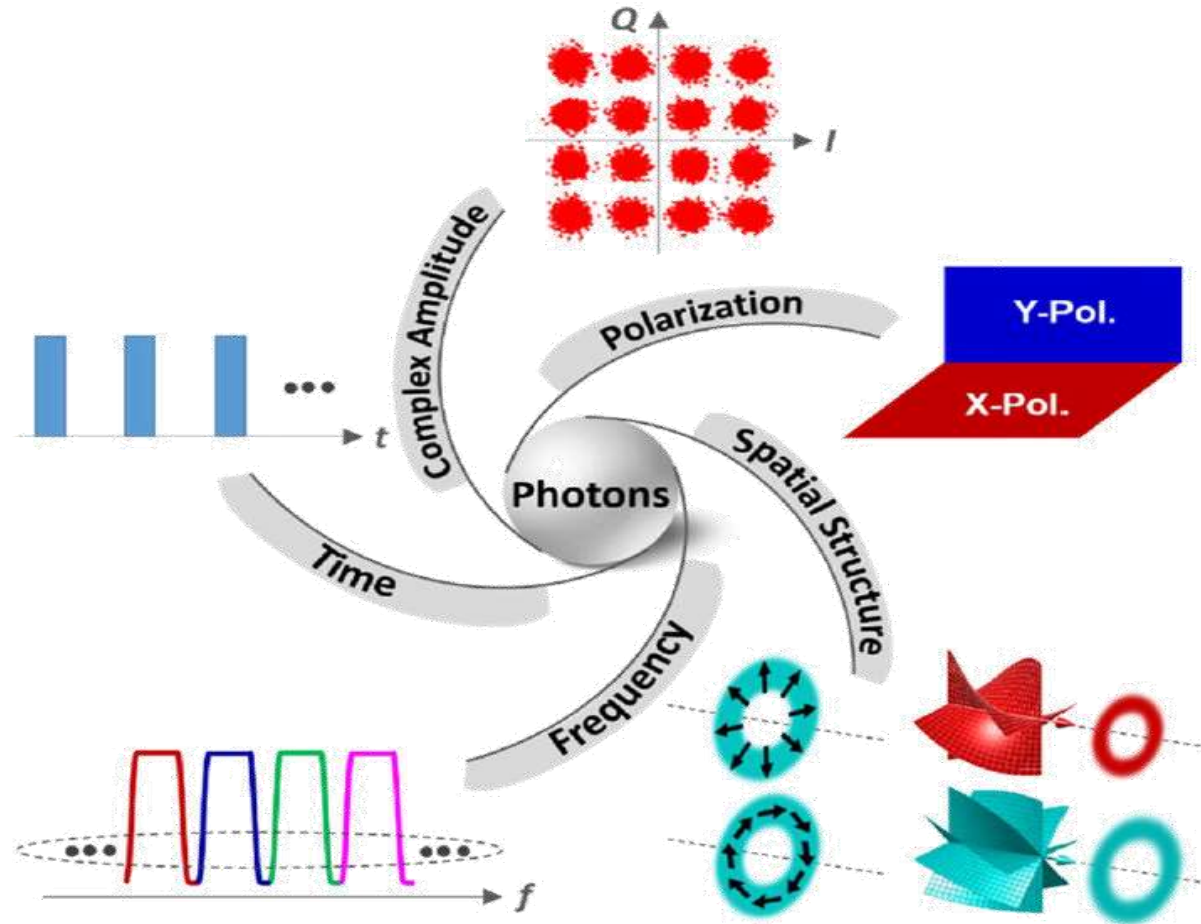


TDM: Time Division Multiplexing mode
PAM: Pulse Amplitude Modulation
PCM: Pulse Code Modulation
LPF: Low Pass Filter

System Diagram of TDM-PCM for Three Analog Signals

7. Brief Introduction to Other Multiplexing Methods

Multiplexing: The technique of transmitting multiple signals over a single communication channel



3. Amplitude Modulation

Summary

